

ECE 3317

Homework #4

Assigned: Wednesday March 25

Due: Monday April 6

Note: In this homework set we are in the sinusoidal steady-state (phasor domain).

- 1) Calculate the input impedance for a transmission line that has a characteristic impedance of $50\ [\Omega]$ that is connected to a load impedance of $100\ [\Omega]$. Assume that the length of the transmission line is $10\ [m]$, and that the transmission line is filled with Teflon, having a relative permittivity of $\epsilon_r = 2.2$. Make a table showing the input impedance for the following frequencies: $0\ [Hz]$, $60\ [Hz]$, $1\ [kHz]$, $1\ [MHz]$, $10\ [MHz]$, $100\ [MHz]$.
- 2) A transmission line is terminated in a normalized load of $Z_L^N = Z_L / Z_0 = 0.5 + j1.0$. Calculate the following:
 - (a) the SWR.
 - (b) the position z_{min} of a voltage minimum (the one that is on the line and closest to the load).
- 3) For the previous problem, make an accurate plot of $|V(z)|/|V^+|$ as a function of z/λ on the line (where z is negative). Plot over the range $-1.0 \leq z/\lambda \leq 0$.

- 1) Calculate the input impedance for a transmission line that has a characteristic impedance of 50 [W] that is connected to a load impedance of 100 [W]. Assume that the length of the transmission line is 10 [m], and that the transmission line is filled with Teflon, having a relative permittivity of $\epsilon_r = 2.2$. Make a table showing the input impedance for the following frequencies: 0 Hz, 60 Hz, 1 [kHz], 1 [MHz], 10 [MHz], 100 [MHz].

$$Z_0 = 50 [\Omega]$$

$$Z_L = 100 [\Omega]$$

$$l = 10 [m]$$

$$\epsilon_r = 2.2$$

$$Z_{in} = Z_0 \left(\frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right)$$

$$\beta = \omega \sqrt{LC} = \omega \sqrt{\mu \epsilon} = \frac{2\pi f}{c_d}$$

$$c_d = 1/\sqrt{LC} = 1/\sqrt{\mu \epsilon} = c/\sqrt{\epsilon_r}$$

$$c = \text{speed of light}$$

$$c_d = \frac{c}{\sqrt{\epsilon_r}} = 2.021 \text{E}8 \quad \beta = \frac{2\pi f}{c_d}$$

frequencies	$\beta l (\text{rad})$	$\tan(\beta l)$	$z_{in} [\Omega]$
0 [Hz]	0	0	100 [Ω]
60 [Hz]	1.865 E-5	1.865 E-5	100 - .00279 j [Ω]
1 [kHz]	3.109 E-4	3.109 E-4	99.99 - 0.0466 j [Ω]
1 [MHz]	0.3109	0.3213	78.079 - 34.111 j [Ω]
10 [MHz]	3.109	- 0.0326	99.682 + 4.87 j [Ω]
100 [MHz]	31.09	- 0.3380	78.478 + 34.797 j [Ω]

2) A transmission line is terminated in a normalized load of $Z_L^N = Z_L / Z_0 = 0.5 + j1.0$. Calculate the following:

(a) the SWR.

(b) the position $z_{\min}$ of a voltage minimum (the one that is on the line and closest to the load).

$$SWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \text{ when it comes to normalised loads } Z_0 = 1$$

$$Z_L = 0.5 + j$$

$$\Gamma_L = \frac{(0.5 + j) - 1}{(0.5 + j) + 1} \Rightarrow \frac{-0.5 + j}{1.5 + j} \cdot \frac{1.5 - j}{1.5 - j} \Rightarrow \frac{-0.75 - 0.5j + 1.5j + 1}{2.25 + 1} \Rightarrow \frac{.25 + 2j}{3.25}$$

$$\Gamma_L = \frac{.25 + 2j}{3.25} \quad |\Gamma_L| = 0.6217$$

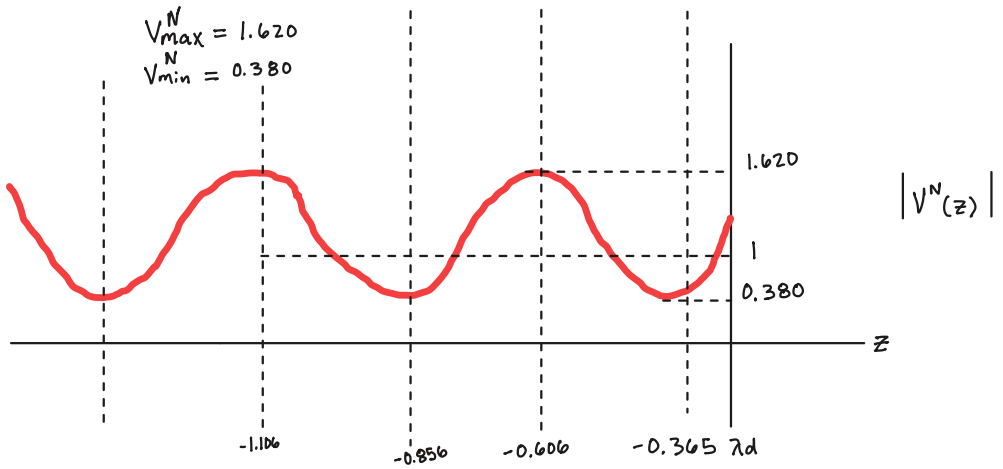
$$a) \quad SWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = 4.255$$

$$b) \quad z_{\min}^v = \frac{-\pi - \phi_L}{2\beta} \quad \phi_L = \angle \Gamma_L$$

$$\phi_L = 1.446 \text{ [rad]}$$

$$z_{\min}^v = \frac{-\pi - 1.446}{2(2\pi/\lambda d)} = -0.365 \lambda d$$

- 3) For the previous problem, make an accurate plot of $|V(z)/|V^+||$ as a function of z/λ on the line (where z is negative). Plot over the range $-1.0 \leq z/\lambda \leq 0$.



- 4) For a normalized load impedance of $Z_L^N = Z_L / Z_0 = 0.6 + j0.4$, find the location of the first voltage maximum and the first voltage minimum from the load end. (The word “first” means the ones that are the closest to the load.) Repeat for the current. Your answers will be in terms of wavelengths (i.e., z/λ).
- 5) A transmission line is short-circuited ($Z_L = 0$).
- Find the expression for $|V(z)|$ and $|I(z)|$ as a function of kz , Z_0 , and V^+ .
 - Sketch $|V(z)|$ and $|I(z)|$.
 - Find VSWR on the line.
- 6) Repeat problem 5 for a transmission line with an open circuit on the load ($Z_L = \infty$).
- 7) Repeat problem 5 for a transmission line with a matched load ($Z_L = Z_0$).
- 8) A transmission line is terminated with a normalized load of $0.8 + j1.0$. Calculate (a) the VSWR, (b) the position of the voltage minimum, and (c) the percentage of the incident power that is reflected by the load. Sketch $|V(z)|$ as a function of z/λ .

- 4) For a normalized load impedance of $Z_L^N = Z_L / Z_0 = 0.6 + j0.4$, find the location of the first voltage maximum and the first voltage minimum from the load end. (The word "first" means the ones that are the closest to the load.) Repeat for the current. Your answers will be in terms of wavelengths (i.e., z/λ).

$$\Gamma_L = \frac{(0.6 + j0.4) - 1}{(0.6 + j0.4) + 1} \Rightarrow \frac{-0.4 + j0.4}{1.6 + j0.4} \cdot \frac{1.6 - j0.4}{1.6 - j0.4} \Rightarrow \frac{-0.64 + 0.16j + 0.64j + 1.6}{2.56 + 1.6}$$

$$\Gamma_L = \frac{-0.48 + 0.8j}{2.72}$$

$$|\Gamma_L| = 0.343 \quad \phi_L = \angle \Gamma_L = 2.111 [\text{rad}]$$

$$\left. \begin{array}{l} V_{\max} \text{ when } \phi_L + 2\beta z_{\max}^v = 0, \pm 2\pi, \dots \\ V_{\min} \text{ when } \phi_L + 2\beta z_{\min}^v = \pm \pi, \pm 3\pi, \dots \end{array} \right\}$$

$$V_{\max} = \frac{0 - 2.111}{2 \left(\frac{2\pi}{\lambda d} \right)} = -0.168 \lambda d$$

$$V_{\min} = \frac{\pi - 2.111}{2 \left(\frac{2\pi}{\lambda d} \right)} = -0.418 \lambda d$$

$$I_{\min} = -0.168 \lambda d$$

$$I_{\max} = -0.418 \lambda d$$

5) A transmission line is short-circuited ($Z_L = 0$).

$$\gamma = \alpha + j\beta \quad (\text{propagation constant})$$

a. Find the expression for $|V(z)|$ and $|I(z)|$ as a function of kz , Z_0 , and V^+ .

$$k_z = \beta - j\alpha \quad (\text{propagation wavenumber})$$

$$\text{Note: } \gamma = jk_z$$

b. Sketch $|V(z)|$ and $|I(z)|$.

c. Find VSWR on the line.

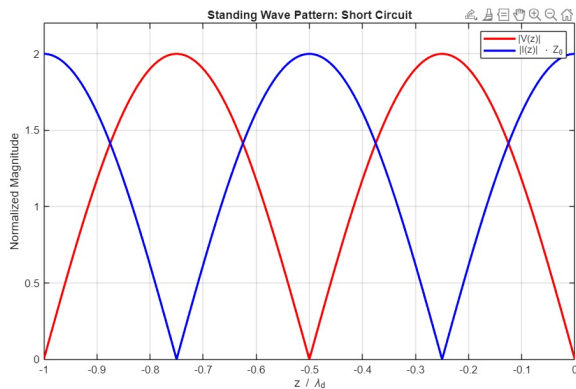
$$V^+(z) = A e^{-\gamma z} = A e^{-jk_z z} = A e^{-\alpha z} e^{-j\beta z}$$

$$j\alpha = 0, \text{ so } k_z = \beta$$

$$a) |V(z)| = V^+ |1 + e^{j(\phi_L + 2k_z z)}|$$

$$|I(z)| = \frac{V^+}{Z_0} |1 - e^{j(\phi_L + 2k_z z)}|$$

b)



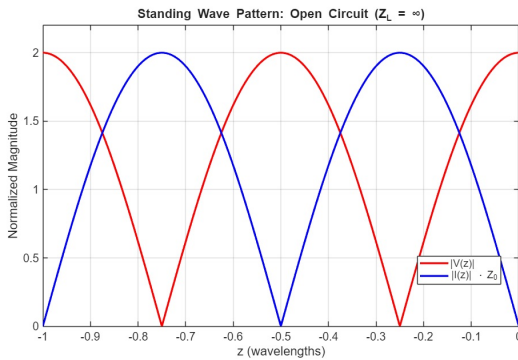
$$c.) \text{VSWR} = \frac{1+1}{1-0} = \infty$$

6) Repeat problem 5 for a transmission line with an open circuit on the load ($Z_L = \infty$).

$$a) |V(z)| = V^+ |1 + e^{j(2kz)}|$$

$$|I(z)| = \frac{V^+}{Z_0} |1 - e^{j(2kz)}|$$

b)



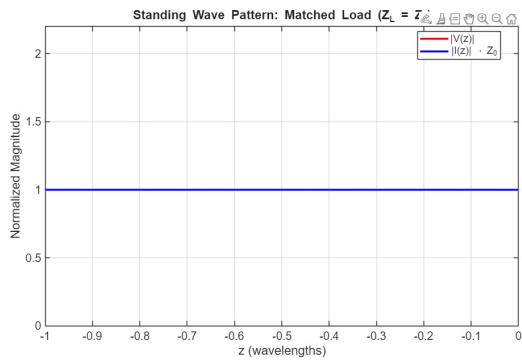
$$c.) VSWR = \frac{1+1}{1-0} = \infty$$

7) Repeat problem 5 for a transmission line with a matched load ($Z_L = Z_0$).

$$a) |V(z)| = V^+$$

$$|I(z)| = \frac{V^+}{Z_0}$$

b)



$$c.) VSWR = \frac{1+0}{1-0} = 1$$

- 8) A transmission line is terminated with a normalized load of $0.8 + j1.0$. Calculate (a) the VSWR, (b) the position of the voltage minimum, and (c) the percentage of the incident power that is reflected by the load. Sketch $|V(z)|$ as a function of z/λ .

$$a) \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(0.8 + j) - 1}{(0.8 + j) + 1} = \frac{-0.2 + j}{1.8 + j} \cdot \frac{1.8 - j}{1.8 - j} = \frac{-0.36 + 0.2j + 1.8j + 1}{3.24 + 1}$$

$$\Gamma_L = \frac{0.64 + 2j}{4.24}$$

$$|\Gamma_L| = 0.4953 \quad \text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$\text{VSWR} = \frac{1 + 0.4953}{1 - 0.4953} = 2.962$$

$$b) \phi_L = 1.261$$

$$z_{\min}^v = \frac{-\pi - 1.261}{2(\frac{2\pi}{\lambda_d})} = -0.3503 \lambda_d$$

$$c) |\Gamma_L|^2 \cdot 100 = 24.53\%$$

